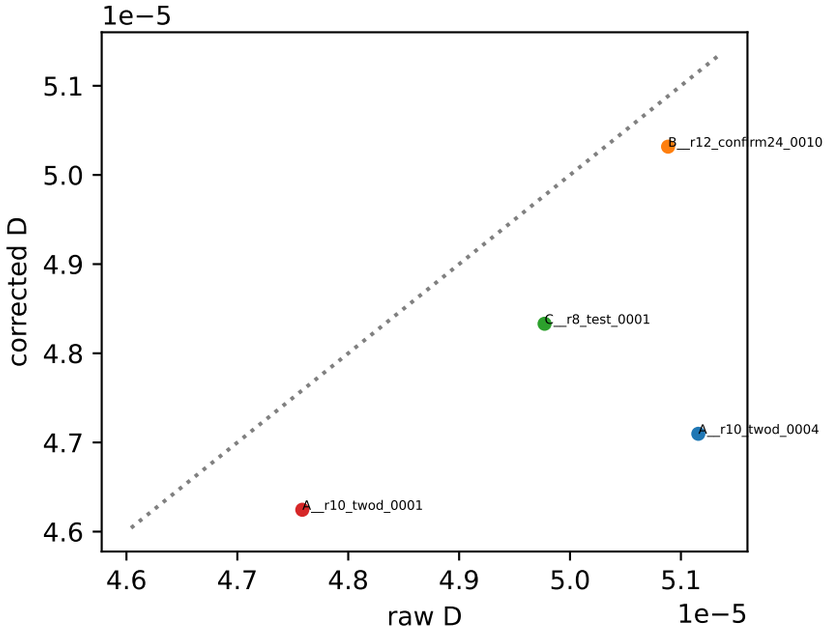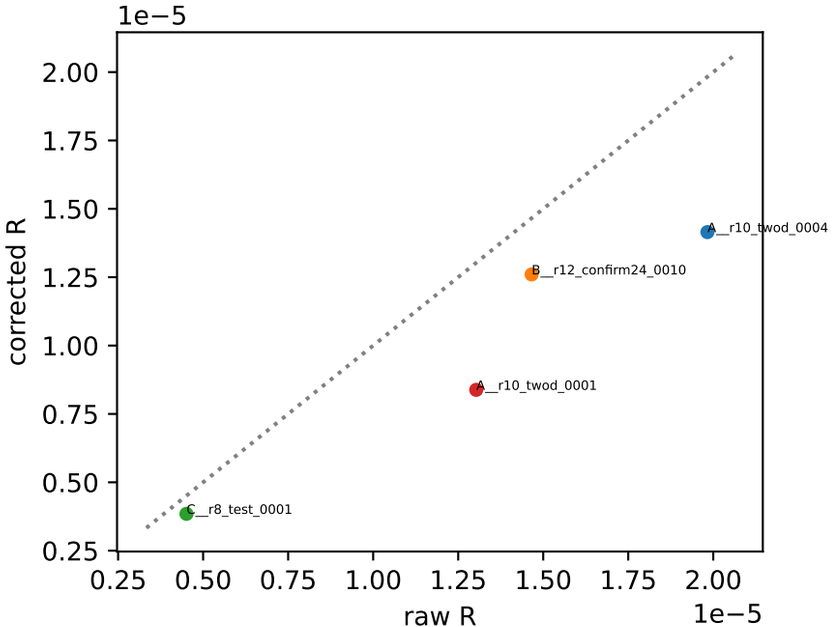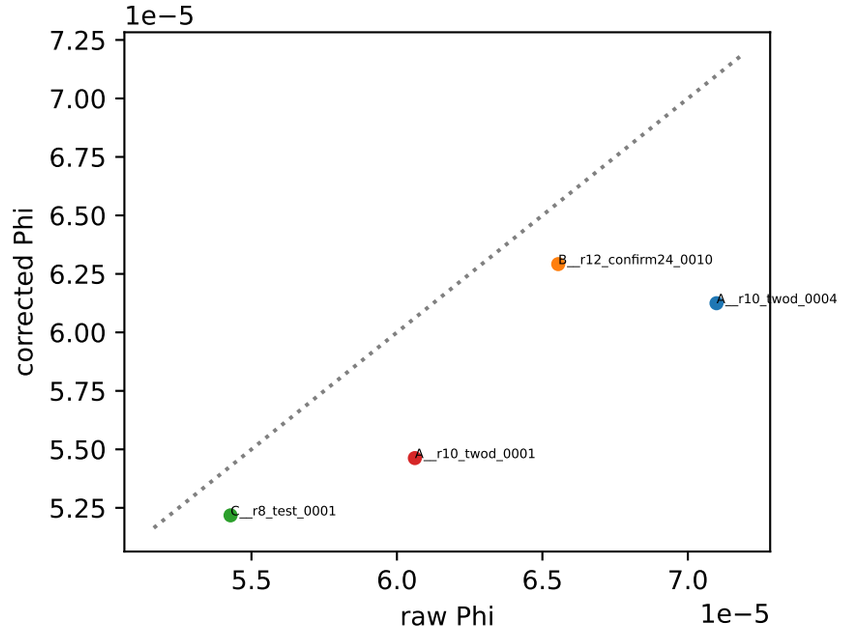online_event corrector decomposition; lower objective need not improve material